Manual

Meantime Report

MDE-MTR 8.2

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Overview

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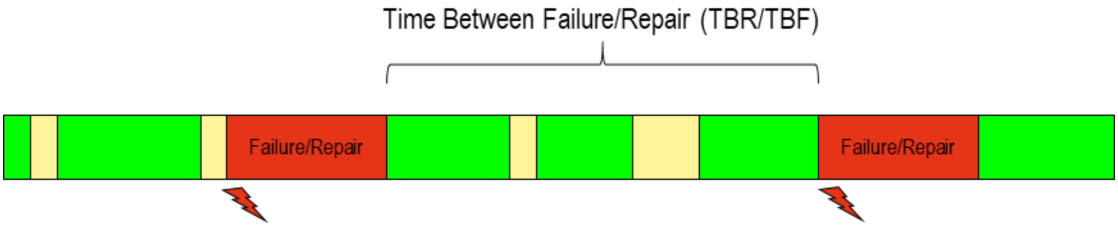
Purpose

The evaluation *Meantime Report* can show the key performance indicators for the average duration (mean time) for machines in a selected period of time. The key performance indicators are calculated using different mean values. The calculation of these mean values is based on MDE postings. For example:

- Mean Time Between Failure (MTBF).
- Mean Time Between Repair (MTBR).
- Mean Time To Repair (MTTR).
- Mean Time To Setup (MTTS).

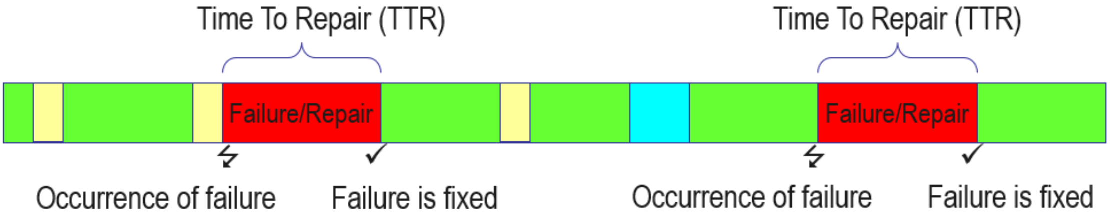
There are two types of key performance indicators:

1. KPIs evaluating times between events (Mean Time Between)



Question: What is the average time that passes between the defined events (failure/repair)?

2. KPIs evaluating the duration of events (Mean Time To)



Question: How long does a defined event take on average (failure/repair)?

The events are concrete (machine) statuses. Note: You can use several (different) statuses to calculate a KPI.

You require two values to calculate the KPIs:

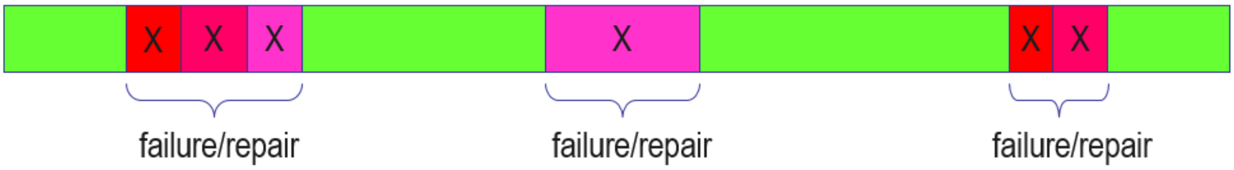
1. Frequency (or number)
2. Duration

For both values, you specify the (machine) statuses in the configuration that are used for the calculation. Indirectly you then also specify the statuses that are not used.

Calculating the frequency

To identify the frequency, the configuration specifies the statuses used. The system counts the statuses (total) that were available in the the evaluation period specified.

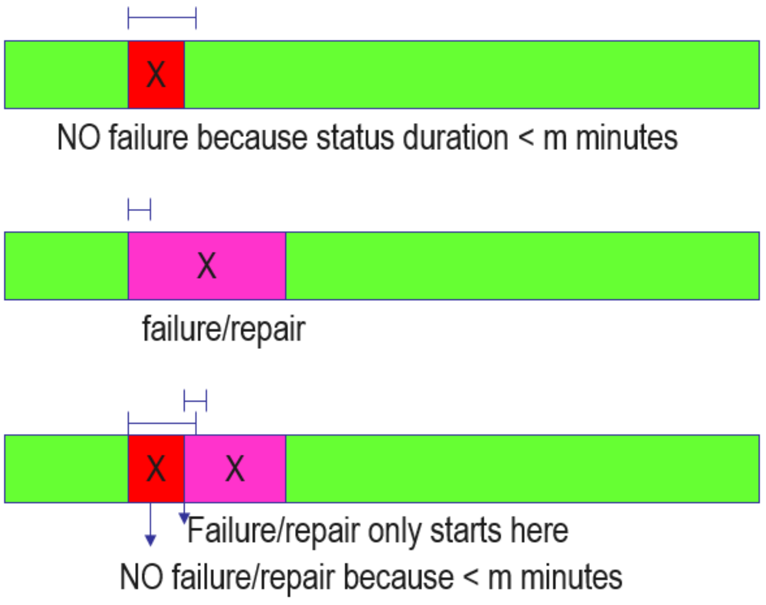
You can specify several statuses in the configuration that are combined to one "event". For example, succeeding statuses can be used to identify a failure:



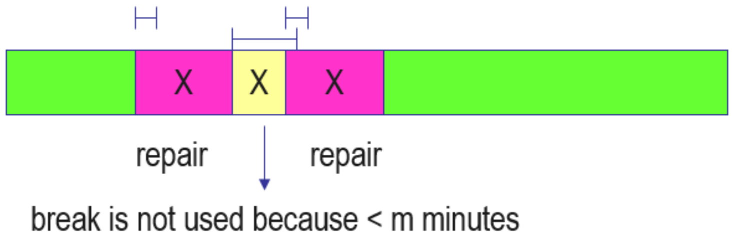
Examples:

- For MTBF you specify which statuses identify a "failure".
- For MTTS you specify the statuses that identify the setup.

When the frequency is identified, only the statuses with a minimum duration m are used (in the diagram this minimum duration is the blue line). You specify the minimum duration for each status. The time of the statuses that are below the minimum duration are not used for calculation.



Example "repair" (MTBR)



In the example above, the break is shorter than the minimum duration specified for the status "break". The "break" status is therefore not used for the calculation and the two statuses "repair" are regarded as two succeeding statuses. The frequency of the event "repair" is then 1.

The breaks are only identified as breaks if an explicit status "break" is available. The breaks that are automatically calculated using the shift model are ignored.

To identify the number (frequency) of failures, only the events are used that started in the evaluation period.

Calculating the duration

To identify the duration, the time between events is used (Time Between...) or the duration is identified using the events (Time to...).

Calculating the time between events - Mean Time Between

These times are required to calculate the KPIs for the mean times between events.

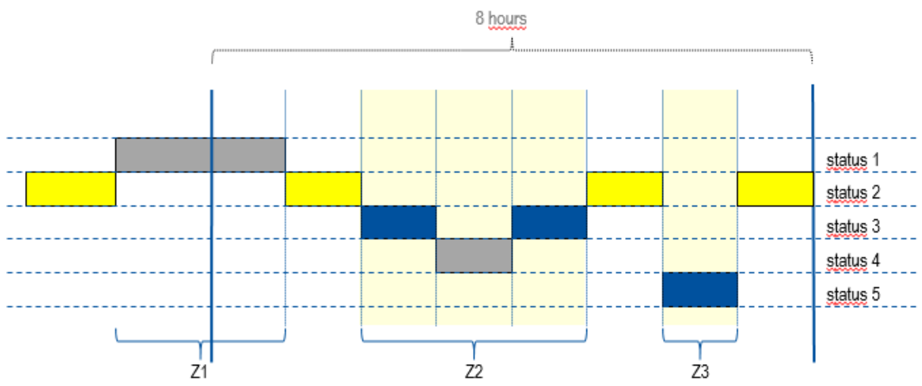
In the configuration, you define the statuses used to calculate the times between the events. With the events, the system then calculates the "time between".

Example with sample configuration:

For the machine 4711, the key figure MTBR is calculated. The status (there can also be several) for the repair is the status 2.
The statuses used to calculate the time between the events are the statuses 3 and 5.
The configuration can be as follows:

Object type	ID1	ID2	ID3	ID4	Function	Value
KPI_MT	MTBR				NAME	Mean Time Between Repair
KPI_MT	MTBR				NAME_SHORT	MTBR
KPI_MT	MTBR				CALCULATION	BETWEEN
KPI_MT	MTBR	MSTAT	4711	2	COUNT	0
KPI_MT	MTBR	MSTAT	4711	3	DURATION	0
KPI_MT	MTBR	MSTAT	4711	5	DURATION	0

1. Example scenario with result:
A shift with a total duration of 8 hours is evaluated.



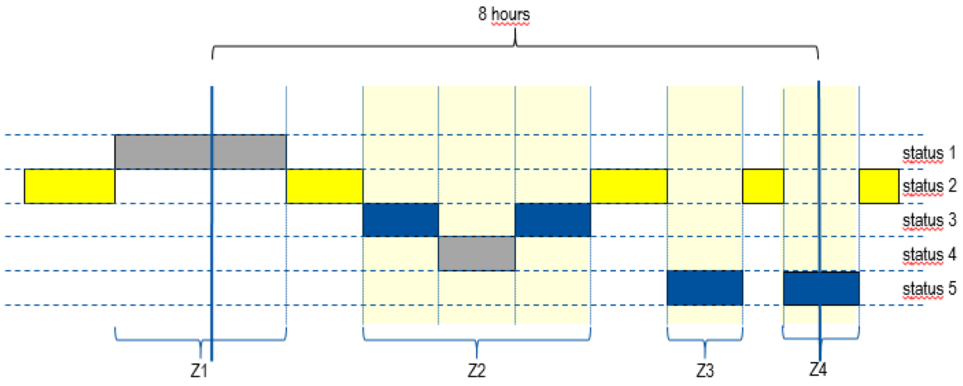
Result:

Machine	...	Short	Designation (name)	Frequency	Duration (total	KPI
4711		MTBR	Mean Time Between Repair	2	3 h	1.5 h

To identify the times, only the events are used that started in the evaluation period. The period Z1 is not used for the KPI calculation of this shift because the status started before the beginning of this shift.

2. Example scenario with result:

A shift with a total duration of 8 hours is evaluated.



<u>Result:</u>

Machine	...	Short	Designation (name)	Frequency	Duration (total	KPI
4711		MTBR	Mean Time Between Repair	3	3.5 h	1.17 h

To identify the times, only the events are used that started in the evaluation period.

Calculation of times using events - Mean Time To

These times are required to calculate the KPIs for the times that the events themselves take. *Question*: How long does a defined event take on average (failure/repair)?

Example with sample configuration:

You want to calculate the KPI Mean Time To Repair <> average repair time (MTTR). The status (there can also be several) for the repair is the status **2.**
With a "Mean Time To" KPI, you must store this status in the configuration in both definitions, with COUNT (frequency) and with DURATION (time).

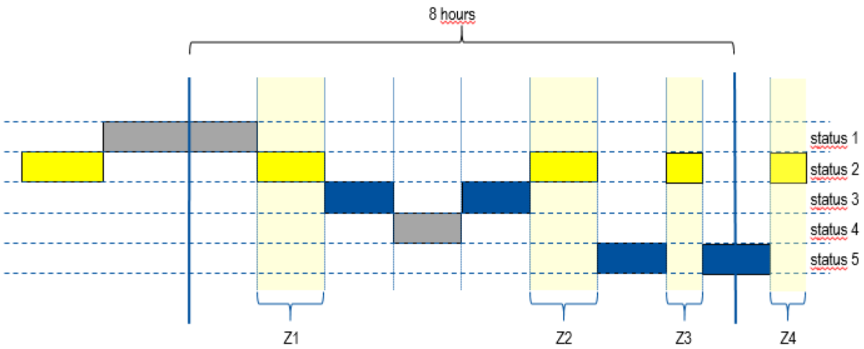
The configuration can be as follows:

Object type	ID1	ID2	ID3	ID4	Function	Value
KPI_MT	MTTR				NAME	Mean Time To Repair

KPI_MT	MTTR				NAME_SHORT	MTTR
KPI_MT	MTTR				CALCULATION	TO
KPI_MT	MTTR	MSTAT	4711	**2**	COUNT	0
KPI_MT	MTTR	MSTAT	4711	**2**	DURATION	0

1. Example scenario with result:

A shift with a total duration of 8 hours is evaluated.



<u>Result:</u>

Machine	…	Short	Designation (name)	Frequency	Duration (total	KPI
4711		MTTR	Mean Time To Repair	3	2.5 h	0.83 h

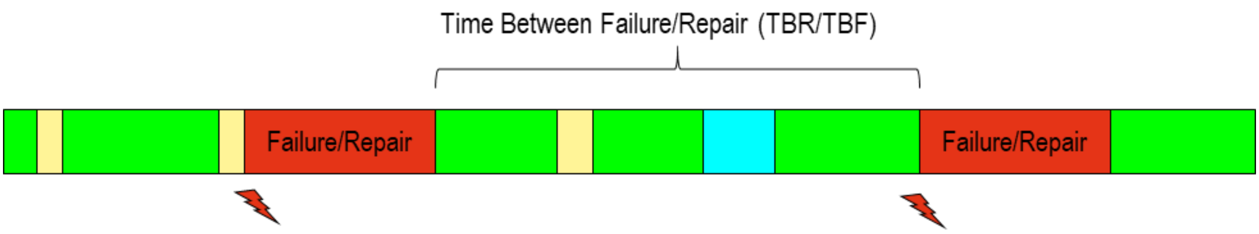
Calculating the KPI

The average time (Mean Time..., e.g. MTBR) is calculated from the (total) time divided by the frequency (number of failure events). If the frequency in the evaluation period is 0, then MTBR is 0.

Definition of terms

MTBR - Mean Time Between Repair

Average time of operation between two failures (and their repair)

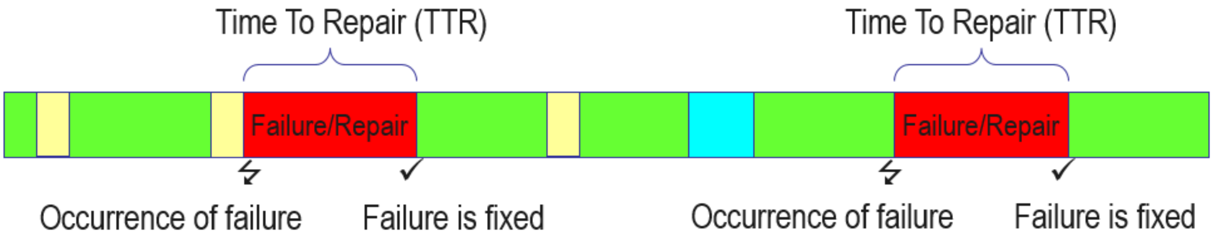


In practice, MTBF is usually defined or calculated as the quotient of the hours of operation (in working order) to the failures identified in the specified evaluation period. It does not matter if the systems can be repaired or not.

If a machine can be repaired, then the machine is repaired when a failure occurs. For this reason, the term MTBR can usually be regarded as a synonym of MTBF with machines. This term is therefore used in the following.

TTR - Time To Repair

Time that passes between the occurrence of a failure and its repair.

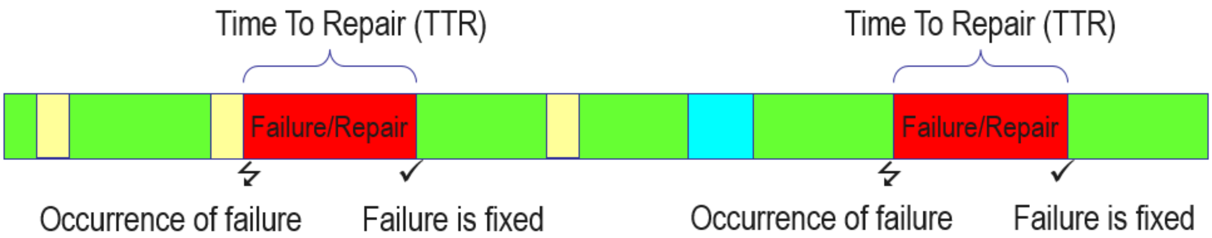


TTTR - Total Time To Repair

Total time of all TTR in a specified period.

MTTR - Mean Time To Repair

Average time that passes between the occurrence of a failure and its repair.



Procedure/processing

To evaluate the data by shift or by time, the data is saved in a separate, aggregated tables (mde_meantime).

For group workplaces, the data is not aggregated. Background: For group workplaces, different statuses are not available. An evaluation based on statuses is therefore not useful.

Selection criteria

The application provides the following selection criteria:

Date from ... to ...

The selection criterion *Date from ... to ...* specifies the period you want to evaluate.

Shift / Time

Use the selection criterion *Shift* and/or *Time* to further narrow down the specified period (date from ... to ...). To do so, select a shift or specify a time (from ... until...).

Machine

The selection criterion *Machine* specifies a workplace that is stored in the machine/workplace master data. You can also use wildcards (placeholders *).

Group

The selection criterion *Group* refers to the group in the machine or workplace master data. The application shows all machines and/or workplaces that are assigned to the selected group. You can also use wildcards (placeholders *).

Cost center

The selection criterion *Cost center* refers to the cost center stored in the machine or workplace master data. The application shows all machines or workplaces that are assigned to the selected cost center. You can also use wildcards (placeholders *).

Company

The selection criterion *Company* refers to the company defined in the machine or workplace master data. The application shows all machines or workplaces that are assigned to the selected company. You can also use wildcards (placeholders *).

Responsibility area

The selection criterion *Responsibility area* refers to the responsibility area defined in the machine master data. Note: A user can only view those machines the user is authorized for (responsiblity area).

Report group

The selection criterion *Report group* refers to the report groups. The application shows all machines or workplaces that are assigned to the selected report group.

Key figure

Narrows down the data to the key figure required (example MTBR for MeanTime Between Repair). The selection provides all values that are configured in the *Advanced object configuration*.

Table view

Master data

The columns correspond to the above selection criteria. The group is the group according to the workplace/machine configuration. The table shows not only the machine like with the selection criteria, but also the designation and the short name of the machine.

Key Figures

Short designation/Designation

The short designation and the designation show the meaning of a key figure according to the configuration.

Frequency

The number specifies how often an event occurred. To calculate the frequency, the number of events in the selected period of time is totaled.

Total time

The *Total time* shows the total time of the combined events (configured as DURATION).

Key figure

The key figure is the average time that the event takes. The key figure is calculated using the frequency/total time.

If the frequency = 1 in the evaluation period, then the average time is the total time.

If the frequency = 0 in the evaluation period, then the average time is 0.

Note: To identify if an event is included in the evaluation period or not, the start time of the event is referenced. With longer failures, this can have the result that the suceeding periods of time cannot be calculated, for example.

The evaluation results by shift or by specified time cannot be compared because the data that is identified and saved can be different according to the aggregation level. It is possible that no result row is displayed for a machine with a selection by time, but with a selection by shift a result row is displayed.

It is recommended to perform this evaluation in general for larger periods of time.

Toolbar

 **Status report**

Function authorization: mstrp

Use this button to call the application Status report.

Transfer of machine number, date and shift or time.

 **Machine time profile**

Function authorization: mtpf

Use this button to call the application Machine time profile.

Transfer of machine number, date and shift or time.

3 Configuration of the Mean Time Report

Purpose

The evaluation *Mean Time Report shows* an average duration (mean time) for machines in a selected period. The basis for the report is cyclically calculated by the system and stored in a separate, aggregated table (mde_meantime).

Please execute the following steps in order for the system to perform an evaluation:

Configuration based on objects

You define the required configuration with the help of an extended object configuration (transaction code adoc). Enter the following in the configuration for the specific machines and KPIs:

Information on a KPI:

Define the information per KPI for the extended object configuration.

Object type: fixed "KPI_MT"(MT=MeanTime)

ID1: KPI Identification (ID), e.g. MTBF, MTTR

ID2: empty

ID3: empty

ID4: empty

Parameter/parameter value:

Parameter	Parameter value	
NAME	Name of the KPI, e.g. Mean Time Between Repair	
NAME_SHORT	Abbreviation of the KPI, e.g. MTBR	
CALCULATION	BETWEEN	for MeanTime **BETWEEN** (e.g. MTBF/MTBR)
	TO	for MeanTime **TO** (e.g. MTTR)

Active: fixed "J"

<u>Information on status assignment</u>

Define the information per machine, KPI and status in the extended object configuration.

Object type: fixed "KPI_MT"(MT=MeanTime)

ID1: KPI Identification (ID), e.g. MTBF, MTTR

ID2: fixed "MSTAT"; set for the status that is included in the calculation for the frequency or duration.

ID3: Machine/workplace number; enter for numeric workstation/machine numbers 8 digits with leading zeros.

ID4: Status number according to configuration

Parameter/parameter value:

Parameter	Parameter value
COUNT	Number If the number of the status is smaller than the value, then the status is ignored. The status is included for 0.
DURATION	Minimum duration in minutes If the duration of the status is smaller than the value, then the status is ignored. The status is included for 0.

Active: fixed "J"

Pre-aggregated data

Store the required transaction data that have been pre-aggregated in a specific table. Data for the evaluation are divided in shift and time. Data is pre-aggregated once a day cyclically.

Machines for which no object-related configuration was performed to calculate the MTBR are not included. Therefore, these machines cannot be evaluated.

Define and activate the configuration for the HYDRA scheduler as follows:

<u>Program start</u>

Hymtrkomp.exe

Parameter to call the program:

- /RECALC=x
  Days in the past in in which the data is (re)determined. Sometimes, existing data in this period is deleted.
  Default: 1 day
  A day is here:
  - for the shift evaluation: a shift day
  - for the time evaluation: the previous day

- /KENNZ=x
  Pre-aggregated data for the KPI
  Default: If not specified, then the data is pre-aggregated for all defined KPIs.

- GRP=x
  Evaluation group for which resources the data is pre-aggregated (individual workplaces). Default: If not specified, all resources are aggregated (individual workplaces).

If you want to update the aggregated data, call the aggregation program again but you need to modify the parameter (RECALC).

Re-organize the data

If data is reorganized, you can set up a time when the data stored in the aggregated table is deleted automatically.

Data is deleted from the aggregated table according to the (default) parameter. The data is not transferred in other tables or exported into files.

The data reorganization is performed in batch operations during a cyclic job. The HYDRA schedulers controls the cyclic job with a separate entry.

Program start

    Hymtrkomp.exe /MOD=DELDATA

Parameter to call the program:

- /DELDAYS=x
  The system deletes all old data.

  Default: 1096 days (= 3 years)

- /KENNZ=x
  KPIs to be deleted.
  Default: all KPIs

- GRP=x
  Evaluation group for which resources the data is deleted (individual workplaces).
  Default: all resources